4(a)	Diffraction is the spreading of waves when they pass through a small opening or around an obstacle.
4(b)	1. The waves from the two sources have to be coherent. 2. The amplitude of both waves have to be equal or approximately equal.
4(c)(i)	$x = \frac{\lambda D}{a}$ gradient of graph $= \frac{(8.8 - 4.4) \times 10^{-3}}{2.0 - 1.0}$ $= 0.0044$ Hence, $\frac{x}{D} = \frac{\lambda}{a} = 0.0044$ $\lambda = 0.0044a$ $= (0.0044)(0.12 \times 10^{-3})$ $= 5.3 \times 10^{-7}$ m
4(c)(ii)	Let the original amplitude of the light from each slit be A. Calculating the resultant amplitude of the bright and dark fringes, $A_{bright} = A + 0.5A = 1.5A$ $A_{dark} = A - 0.5A = 0.5A$ $\frac{A_{bright}}{A_{dark}} = \frac{1.5A}{0.5A} = 3.0$ Since $I \propto A^2$, $\frac{I_{bright}}{I_{dark}} = \left(\frac{A_{bright}}{A_{dark}}\right)^2 = 3.0^2 = 9.0$

No.	Solution
5(a)	